



Dental consideration of Psychiatric disorders

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Outlines

- **By the end of this presentation, you should be able to:**

- **Define mental disorders.**
- **Classify mental disorders.**
- **Understand how psychiatric disorders influence the quality of life.**
- **Discuss appropriate management needs for persons with mental disorder.**



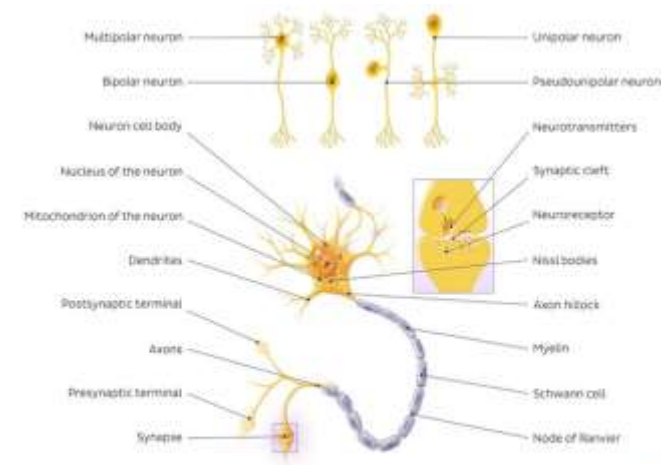
Definition of mental disorders

- **Mental illness or disorders** are characterized by a clinically significant **disturbance in an individual's** cognition, **thinking**, emotion regulation, **mood or behavior** (one or combination).
- **Associated with:**
 - **Distress and/or**
 - **Impairment in functioning** (social, work or family activities).
 - **Risk of self-harm** (Substance abuse or suicide).
- In 2019, anxiety (feature of modern life) and depression (affects about 264 million people worldwide) are the most common mental disorders.



Causes and risk factors

- Genetic or positive family history (depression, schizophrenia and anxiety).
- Developmental:
 - **Abnormal balance of neurotransmitters** {chemical messengers that transmit a message from a nerve cell across the synapse to a target cell (another nerve cell, or a muscle cell, or a gland cell)}.
 - **Defects or injury** to certain areas of the brain.



- **Environmental or Psychological trauma:**
 - **Neglect, poverty, violence, disability.**
 - **Socioeconomic inequality.**
 - **Changing jobs/schools or chronic medical disease.**
 - **Severe emotion, stress, sexual or drug abuse (cannabis, alcohol and caffeine.**

What dental professionals should know about psychiatric disorders

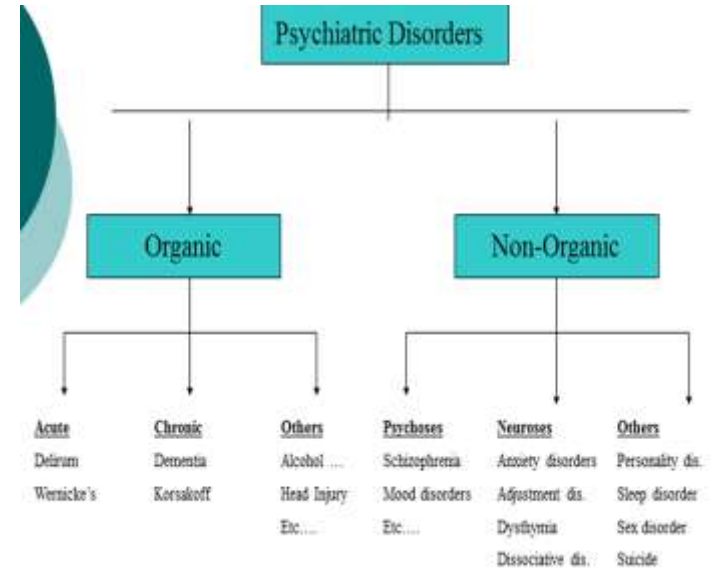
- Psychiatric disorders are common and prevalent, often under-recognized and underdiagnosed.
- These can be associated with medical co morbidity and increased dental problems.
- Psychiatric illness may present with physical symptoms as pain.
- These also associated with higher rates of substance use disorders.
- Medication of psychiatric illness may have oral adverse effects, also may interact with drugs used in dentistry.

- **People with mental disorders (type and severity) increase risk of oral diseases (dental caries and periodontal disease) because:**
 - **Lack of personal perception of oral health and its problems.**
 - **Self-induced lesions and neglect oral hygiene.**
 - **Side effects of psychiatric drugs.**

- **Also, dental treatment is difficult for these patients because:**
 - **Compliance with appointments or treatment is often poor.**
 - **Difficulties in gaining informed consent to treatment.**
 - **Lack of motivation and apathy.**
 - **Unable to cooperate, difficult mobility and fear of treatment.**
 - **Poor communication as well as financial considerations.**

Features suggesting of organic mental illness

- **Disturbed consciousness.**
- **Disturbed cognitive functions:**
 - a. **Attention and concentration.**
 - b. **Orientation:** time, place & person.
 - c. **Memory:** immediate, recent and remote.
- **Presence of chronic disease** e.g. diabetes mellitus (DM), hypertension (HTN).
- **Presence of neurological features** e.g. dysarthria & ataxia.
- **Old age** onset.



Schizophrenia

- It is characterized by significant impairments in thoughts, perceptions and changes in behavior **with** loss of contact with reality.
- Intelligence is unimpaired and patients may be aware that their behavior is bizarre.
- Diagnosis by presence of **persistent** at least **2 symptoms for 6 months:**
 - **Disorders of thought (delusions)** (false beliefs not subject to reason or evidence and are not explained by a person's usual cultural concepts).

- **Disorders of perception (hallucinations)** (visual or hearing).
- **Disorganized speech** (speak incoherently, respond to questions with unrelated answers).
 - Signs of disorganized speech involve:
 - **Loose associations:** Rapidly shifting between topics with no connections between topics.
 - **Perseveration:** Repeating the same things repeatedly.
 - **When cognitive disorganization is severe,** it nearly impossible to understand what the person is saying.

- **Disorganized or catatonic behavior or extreme agitation (bizarre behavior).** It may include:
 - Inappropriate emotional responses (frightened, anxious and confused).
 - Routine behaviors such as bathing, dressing, or brushing teeth can be severely impaired or lost.



Treatment of schizophrenia

- **Early active intervention before the illness becomes chronic.**
- **Assessment of the risk to themselves or others.**
- **Family therapy.**
- **Cognitive behavioral therapy (CBT).**
- **Long-term psychotherapy and antipsychotic drug.**
- **Supportive therapy** via community psychiatric services.
- **A person with schizophrenia can be sensitive to stress, so best avoid it and keep calm.**

Antipsychotic Medications

- 1st generation Antipsychotics (conventional):
 - **Chlorpromazine** (low potency)
 - Haloperidol (high potency):
 - Oral side effects:
 - Xerostomia, facial dyskinesia involve **bulbar or neck muscles** with subsequent difficulties in speech or swallowing.

- **Other:** orthostatic hypotension.
 - **Treatment:** Patients should be **raised slowly from dental chair.**
- **2nd generation Antipsychotics (atypical):**
 - **Clozapine, risperidone.**
 - » **Oral side effects:**
 - **Xerostomia, dysphagia, stomatitis.**

Dental aspect of schizophrenia

- Schizophrenic peoples are at greater risk of:
 - Dental caries, gingivitis/advanced periodontal disease, tooth loss, poor oral hygiene, mucosal diseases... +
 - Poor dietary habits, smoking, alcohol abuse, substance abuse...
 - Higher prevalence of severe tooth damage.



- **Conscious sedation is safe.**
- **Schizophrenia-medication side effects:**
 - **Haloperidol and phenothiazines may cause orthostatic hypotension.**
 - **Patients should be raised slowly from the dental chair.**
 - **Haloperidol block the vasoconstrictor activity of adrenaline (epinephrine).**

– **Haloperidol and clozapine** can cause **hypersalivation**.

– **With tramadol**, there is a **risk of seizures**.

- **General anesthesia (GA)**, can lead to **severe hypotension**, so should be avoided if possible.
- **Neuroleptics** may intensify effects of sedatives, hypnotics, opioids, antihistamines leading to **Severe respiratory depression**.

- **Neuroleptics** can **cause**:
 - **Xerostomia** (with susceptibility to candidiasis and caries, and parotitis) and oral pigmentation.
 - Muscular rigidity (facial dyskinesias), involve bulbar or neck muscles, with subsequent difficulties in speech or swallowing. Alternatively, there may be uncontrollable facial grimacing.

- **Oral dyskinesia or Tardive dyskinesia (TD):**
 - It affects about 15–20% of patients who have been receiving **phenothiazines (Chlorpromazine)**.
 - **Abnormal involuntary**, purposeless movements involve mouth, **tongue, lips, jaws** (can extend to trunk/limbs).
 - **Causes:**
 - Prolonged use or high – dose **antipsychotics**.
 - Neuropsychiatric conditions.



Tardive dyskinesia

- Protrusion and rolling the tongue
- Sucking and smacking movements of the lips
- Chewing motion
- Facial dyskinesia
- Involuntary movements of the body and extremities

- **Complication of oral dyskinesia:**

- Tooth wear, oral pain/injury.
- Temporomandibular joint (TMJ) degeneration.
- Impaired speech, **chewing difficulties**
- Inadequate food intake...wt. loss.
- Loss of social contact



- **Treatment:**

- TD does not respond to withdrawal of the causative drug.
- **Reduce** dose.
- **Switch to another** antipsychotic as clozapine

Depression

- It is a common emotional disorder.
- **Depression** is an **illness**, involve mood and thoughts, affects persons eat and sleep, feel about themselves.
- It is the risk factor for:
 - Alcohol, tobacco or drug abuse (1/3 of patients).
 - Sleep disturbance.
 - Physical inactivity and obesity.
 - Suicide.
- Depression is the 4th leading cause of disability.



- **Risk factors:**

- **Depression more common in females due to:**

- Hormonal factors, particularly **menstrual cycle changes**, **pregnancy**, miscarriage, **pre-menopause and menopause**.
 - It may be genetics and run in family.

- **Episodes triggered by:**

- **Serious and chronic medical illnesses** such as **stroke**, a **heart attack**, cancer.
 - Viral infections, like influenza.
 - Difficult relationships.
 - **Financial problems or** any stressful change in their life.

Major depression

- **Persistent** sad, irritable or anxious, empty.
- **Loss of interest or pleasure** in activities that were once enjoyed for most of the day at least 2 weeks.
- **Other symptoms (at least 4) of the following:**
 - Appetite or weight changes (loss or gain), fatigue.
 - **Loss of self-confidence, decreased energy, loss of libido.**
 - Difficulty making decisions
 - Difficulty thinking, remembering or concentrating.

– Feelings of excessive guilt or worthlessness, **helplessness** and **hopelessness** about the future.

– **Disturbed sleep** (insomnia, early-morning awakening or oversleeping).

– **Thoughts of death/ suicide.**



Management of depression

- **History (symptoms).**
- **Treatment:**
 - **Psychotherapy and medication.**
 - **Psychotherapy:**
 - Psychiatrists are required to assess patient's capacity to decide; also, to assess risk for suicide, self-neglect and danger to others.
 - Psychological techniques (cognitive and supportive therapy).
 - Electroconvulsive therapy (ECT) for severe cases {physical retardation or delusions} are present.

- **Monoamine oxidase inhibitors (MAOIs):**

- It is **the 1st effective antidepressants** but now rarely prescribed because of their adverse effects:

- **Dry mouth**
 - **Hypotension, Sexual dysfunction.**
 - **Gastrointestinal:** nausea, constipation (bran, fruit and vegetables).
 - **Interaction with foods** cheese, chocolate, noradrenaline (epinephrine can be used safely), ephedrine causing hypertensive crises.
 - **Enhance respiratory depressants** with general anesthetic agents, sedatives, antihistamines, opioids (seizures with tramadol).

- **Tricyclic antidepressants (TCAs):**
 - **Side effects:**
 - **Xerostomia**, Constipation, Sexual dysfunction.
 - **Blurred vision, dizziness** (rising from the bed or chair slowly).
 - **Drowsiness** (take at bedtime, should not drive)
 - **Potentiate vasoconstrictor effects of epinephrine** in local anesthetics.
 - **Contraindicated in** patients with cardiac disorders (recent myocardial infarct, cardiac failure), epilepsy, liver dysfunction, bleeding tendency, glaucoma and urinary obstruction (prostatic hypertrophy).

- **Selective serotonin reuptake inhibitors (SSRIs):**
 - **Side effects:**
 - **Gastrointestinal:** anorexia, nausea, diarrhea
 - **Xerostomia**, stomatitis, glossitis, bruxism
 - Sexual dysfunction.
- **Antidepressant/Mood Stabilizers:**
 - **Lithium:**
 - **Oral side effects:** **xerostomia** (predispose to oral **candidiasis** and accelerated caries) , lichenoid stomatitis, metallic taste.

Dental aspect of Depression

- Dental treatment is deferred until depression is under control, but preventive programs should be instituted at an early stage.
- Xerostomia (dry mouth) is the most common oral complaint, that cause:
 - Oral candidiasis.
 - Accelerate dental caries.
 - Impaired taste sensation.

Complication of xerostomia

- Increased incidence of caries.
- Oral ulcerations, halitosis.
- Altered taste sensation.
- Difficulties eating, speaking and wearing dentures.
- Decrease amount of saliva increase infections (viral, bacterial, fungal), dysphagia, ease of trauma to oral mucosa, gingivitis & periodontitis.
- Atrophic, fissured or burning tongue.



– Treated by:

- Enhance salivary flow by sugar free mints, chew sugarless, and water.
- Clean teeth daily by soft brush and avoid brushing immediately after episodes of vomiting.
- Mouth wash with alkaline mouth rinse immediately after vomiting (NaHCO_3), water.

- **Tricyclics and MAOIs** can cause postural hypotension, so dental chair should be brought upright slowly.
- **Bruxism (or facial dyskinesias):** clenching, grinding of the teeth or jaw.
- **Periodontitis**
- **Depression is associated with the following:**
 - **Chronic facial pain, burning mouth or sore tongue.**
 - **Temporomandibular pain dysfunction syndrome.**
 - **Other oral complaints (delusional) include:**
 - **Discharges (of fluid or powder coming into the mouth)**
 - **Dry mouth or sialorrhea despite normal salivary flow.**
 - **Halitosis with impaired taste sensation**

SIX LINKS BETWEEN DEPRESSION AND ORAL HEALTH



Depression is a life-threatening disease linked to a range of health issues. Studies increasingly show it is also linked to poor oral health.

Depression can lead to:



Neglected oral hygiene



Poor nutrition



Drug-induced xerostomia (dry-mouth)



Avoidance of necessary dental care



Increased risk of dangerous dental behaviors



1

61% of patients reporting depression also reported an aching mouth and more than half consider their teeth to be in fair or poor condition.



2

Depression increases the circulating level of cortisol, thus raising the risk that periodontal disease will progress.



3

Depression can lead to temporomandibular disorders that cause pain and restrict jaw movement.



4

Depression is partly an "inflammatory" disorder, meaning that inflammation can induce it. Periodontitis causes such inflammation.



5

Side effects of medicines for depression can increase the risk for caries, periodontal disease, and oral infections. Other side effects may cause tooth damage.



6

Depression is linked to increased high-risk behaviour (e.g. smoking, alcohol and drug abuse) associated with oral damage.

- What is most common dental complication of schizophrenia?
- a) Xerostomia.
- b) Dysphagia.
- c) Periodontal disease.
- d) Stomatitis.

- Which of the following drug cause oral dyskinesia?
- A. Paracetamol.
- B. Antihistaminic.
- C. Lidocaine.
- D. Antipsychotics.

- **How to diagnose Schizophrenia**

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- **List dental aspect of Schizophrenia**

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- **List dental aspect of depression**

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- **Risk factors of depression**

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- **Mention Complication of xerostomia**

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Thank You